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Progress report #2
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Introduction

For the Progress Report 2, I designed the LAN for Company A using Cisco Packet Tracer. The LAN topology used the major network devices such as the router, switches, DMZ switch, internal switch, and wireless access point. According to the project requirements, only one device is used to represent a subnet, and I put the notes to show the total number of devices on a subnet.

2. SUBNET ALLOCATION

In Progress Report 1, I used Sales as a subnet /27. In Progress Report 2, I corrected the subnet of Sales to /26 because the project stated that each router interface should have an IP address in a subnet. The subnet of Sales required an increase in the number of usable addresses. The other subnets were also shifted so they don't overlap.

Subnet	Network ID	Subnet Mask	CIDR	Broadcast	First Host	Last Host	Max Hosts	Default Gateway
Engineering	216.244.168.0	255.255.255.128	/25	216.244.168.127	216.244.168.1	216.244.168.126	126	216.244.168.1
Manufacturing	216.244.168.128	255.255.255.128	/25	216.244.168.255	216.244.168.129	216.244.168.254	126	216.244.168.129
Sales	216.244.169.0	255.255.255.192	/26	216.244.169.63	216.244.169.1	216.244.169.62	62	216.244.169.1
Administration	216.244.169.64	255.255.255.224	/27	216.244.169.95	216.244.169.65	216.244.169.94	30	216.244.169.65
Internal Server	216.244.169.96	255.255.255.240	/28	216.244.169.111	216.244.169.97	216.244.169.110	14	216.244.169.97
DMZ	216.244.169.112	255.255.255.248	/29	216.244.169.119	216.244.169.113	216.244.169.118	6	216.244.169.113
Guest WLAN	216.244.169.	255.255.255.	/26	216.244.169	216.244.169.	216.244.169	62	216.244.169.

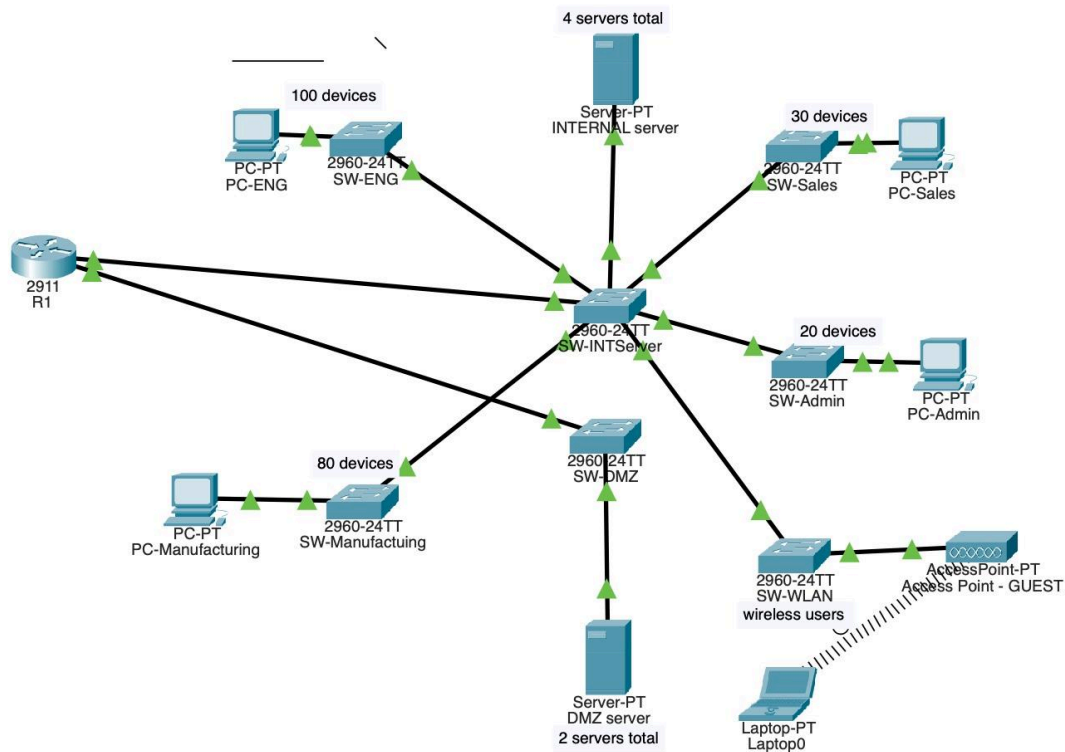
	128	192		.191	129	.190		129
WAN	216.244.175.252	255.255.255.252	/30	216.244.175.255	216.244.175.253	216.244.175.254	2	216.244.175.253

3.CISCO TOPOLOGY:

In Cisco Packet Tracer, the LAN for Company A was created by utilizing one router, several switches, one switch for the internal side, one switch for the DMZ, and one access point for the wireless side. According to the project requirements, only one representative device per subnet is connected, along with a note indicating the total number of devices connected to the subnet. The requirements for the project are that a switch has 24 Ethernet ports, a router has 3 Ethernet ports, and the access point has the capability to support up to 64 users through the wireless side.

The representative devices connected are:

- 1 PC for Engineering
- 1 PC for Manufacturing
- 1 PC for Sales/Marketing
- 1 PC for Administration
- 1 Internal Server
- 1 DMZ Server
- 1 wireless laptop for the Guest WLAN



4.IP Configuration Used in the Topology

The representative devices were configured with IP addresses from their own subnets. The DMZ server and the Internal Server were also tested by pinging their router interfaces, and both tests were successful. This proved that the two subnet connections were properly configured in the current topology.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 216.244.169.97

Pinging 216.244.169.97 with 32 bytes of data:

Reply from 216.244.169.97: bytes=32 time=1ms TTL=255
Reply from 216.244.169.97: bytes=32 time<1ms TTL=255
Reply from 216.244.169.97: bytes=32 time<1ms TTL=255
Reply from 216.244.169.97: bytes=32 time<1ms TTL=255

Ping statistics for 216.244.169.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

INTERNAL SERVER

```
C:\>ping 216.244.169.113

Pinging 216.244.169.113 with 32 bytes of data:

Reply from 216.244.169.113: bytes=32 time=1ms TTL=255
Reply from 216.244.169.113: bytes=32 time=4ms TTL=255
Reply from 216.244.169.113: bytes=32 time=4ms TTL=255
Reply from 216.244.169.113: bytes=32 time=4ms TTL=255

Ping statistics for 216.244.169.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 3ms
```

DMZ SERVER

5.Future Configuration

At this point, the Packet Tracer topology shows the correct subnet design and device placement for Company A. The DMZ subnet and the Internal Server subnet have already been correctly configured and tested successfully. Now for the remaining subnets for the internal departments, the plan will be to utilize VLANs and router subinterfaces so that the Engineering, Manufacturing, Sales/Marketing, Administration department, and the Guest WLAN will each have their own subnet, yet utilize the limited ports available on the router efficiently, as the project assumes the router has only 3 Ethernet ports, so VLANs will be used to efficiently complete the remaining internal routing in the next stage

6.Conclusion

In this progress report, I have configured the LAN of Company A using Cisco Packet Tracer. I have also modified the subnet plan accordingly and added the default gateways to each subnet. In addition, I have created a topology with required devices and end devices. I have also added notes to represent the total number of devices used in each subnet. The DMZ subnet and Internal Server subnet configurations are completed and tested successfully. The other internal subnets will be configured using VLAN technology.